Certified Control of the Full 600-Cell

Orbit structure, constructive algorithms, and a validated native workbench

C600 Studio project

Expanded technical report, September 2026

Abstract

The full-cut 600-cell is a four-dimensional twisty puzzle with 177,120 surface pieces, 259,800 sticker slots, and 35 moving orbits. This report reconstructs the constructive method and software development documented in a 52-page design transcript, its accompanying algorithm reference, and subsequent C600 Studio 0.2.4 validation. The mathematical analysis formalizes piece/frame actions, abelian orientation invariants, guarded setup graphs, pure-star generation, and collateral-preserving stage correctness. Nine independently checked controllers yield more than 10^{151851} face-color states, exceeding an upper bound for the documented solved 4^7 hypercube by over 86,000 decimal orders of magnitude. In a specified 1,800-symbol primitive alphabet, a word-counting argument also gives a worst-case distance of at least 46,648 turns and quantifies why 1,000-step scrambles are far from uniform random states. These results are distinguished from human difficulty, optimal solutions for particular scrambles, and an exact full-group order. Three synthetic compact histories restore every label while representing approximately 353 million primitives each. The engineering analysis explains native generator equivalence, transactional previews and journals, coherent snapshots, exact filtering and picking, bounded log interchange, process ownership, and graphics recovery. Recorded Windows validation supports the frozen native workbench; the expanded report adds explicit source contracts and reproducible arithmetic rather than new graphics benchmarks. Intel HD Graphics 620 measurements support adaptive interaction, while no minimum GPU has yet qualified for continuous full-detail rotation at 1920×1080 and 30 FPS. The contribution is a reproducible computational framework and a detailed research instrument, without asserting a human solving record or universal difficulty ranking.

Keywords: 600-cell; permutation puzzles; computational group theory; nonabelian orientation; certified macros; reproducibility; human-computer interaction.

1 Scale and significance

The number 600 describes the tetrahedral cells of the underlying regular polytope, not the number of pieces requiring control. At the retained cuts of ± 0.968, the full puzzle contains 433 sticker regions per cell, 177,120 physical surface pieces, and 35 moving position orbits. Its 600 fixed single-sticker centers account for the difference between 177,120 total pieces and 176,520 moving pieces. Andrey Astrelin's original announcement already identified the 259,800-sticker scale. The present construction supplies the finer orbit-level machinery needed to work with that scale [1, 2, 3].

The increase is particularly substantial relative to the simplified 600-cell. That subsystem retains only three moving families, together with the fixed centers: 2,640 total pieces and 9,000 stickers. Restoring the omitted pieces adds 174,480 pieces in 32 moving orbits. The ratios are approximately 67.09 in total pieces and 28.87 in stickers. A successful simplified solve therefore provides a valuable structural starting point while leaving most of the full puzzle to be controlled [1, 2, 4].

The strongest defensible statement about orders of magnitude concerns configurations, rather than a universal measure of difficulty. Section 4 constructs at least 10^{151851} distinct face-color states using only nine of the full puzzle's orbits. For comparison, a 4^7 hypercube has 57,344 stickers and 14 face colors. Even allowing every balanced color assignment, including illegal ones, gives fewer than 10^{65696} states. The conservative ratio is therefore greater than 10^{86155}. The 4^7 is a meaningful high-dimensional comparator because a completed solve is recorded in the original MC7D Hall of Fame [5].

This comparison does not make the full 600-cell the largest solved-or-unsolved puzzle under every definition. The same Hall of Fame includes the 3^7, whose 10,206 stickers provide another dimensional benchmark, but a documented team solve of a 256^3 cube involves 393,216 stickers, more than the full 600-cell [5, 6]. Dimensions, sticker counts, reachable configurations, recognition demands, and solution lengths are separate quantities. The number of colors also matters: throughout this report, a color means a distinct solved cell or face symbol, not a claim that a display can make hundreds of RGB shades equally easy to recognize.

Table 1: Scale of the named puzzle comparisons and the evidence used for each status.

Puzzle or model	Sticker slots	Status used for comparison	Interpretation
Simplified 600-cell	9,000	Published completed solve	Three moving families; same underlying cell geometry
3^7 hypercube	10,206	Original MC7D solve records	High-dimensional benchmark
4^7 hypercube	57,344	Original MC7D solve record	Named comparator for the color-state upper bound
Full 600-cell	259,800	Constructive model solutions and validated workbench	Thirty-five moving orbits; no human record asserted here
256^3 cube	393,216	Published participant account of a team solve	Counterexample to a universal sticker-count ranking

Solving this puzzle is useful as a demanding test of constructive control and verifiable computation. A workable method must coordinate many move orbits, distinguish piece identity from appearance, represent noncommuting orientation changes, and preserve already completed work. It also exposes an engineering problem with wider relevance: a system must execute compressed operations quickly while retaining an independently checkable account of everything those operations do. The scientific value lies in explicit constructions, failure analysis, reproducible data, and precise evidence boundaries. Large configuration counts alone neither prove a lower bound on human effort nor establish a new complexity class.

2 Documentary basis and antecedents

2.1 Primary corpus and method

The principal documentary source is *600 Cell Solving Strategy*, a 52-page exported design transcript [1]. Its first 18 pages develop the full model and constructive solving method; pages 19–24 examine a simplified solve; pages 25–31 explain the algorithm reference, replay records, and workload; pages 32–52 develop and debug the graphical workbench. The accompanying *Full 600-Cell: 35-Orbit Algorithms* supplies one algorithm card for every moving orbit, exact words, orientation procedures, and a workload analysis [2].

The transcript is treated as a record of claims and design decisions, not as proof merely because an assertion appears in it. The analysis separates mathematical arguments, finite computational certificates, historical execution reports, synthetic integration tests, and subsequent native Windows checks. A new audit verifies the nine controllers used in the configuration lower bound. The public companion preserves synthetic replay data and the original algorithm reference, while a source guide maps the report's discussion back to transcript page ranges. The original conversation export, account-bearing links, personal sessions, and third-party solve histories are not included in the public research bundle.

The companion *Puzzle Theory of the Full 600-Cell* develops the geometric foundations and optimization conditions in fifteen lemmas and a combined theorem. Its mathematical blueprint passed an independent, informal Rethlas review using a documented Windows adaptation, with no critical errors or gaps. That verdict concerns the stated conditional mathematics, not the entire software stack or this report [15].

2.2 Prior solving and software foundations

Magic Puzzle Ultimate, including its original puzzle simulator and renderer, is Andrey Astrelin's work. C600 Studio is a modification built around that foundation. Its contribution consists of an authoritative full-state engine, certified solving tools, a journal, and adaptations around the inherited native viewport; it does not replace the original authorship claim [3, 7].

Nan Ma's simplified 600-cell solve provides methodological background. His account solves faces, then edges, then corners, using the spherical projection's surface as a buffer region. A piece travels across that surface to a point above its destination, is inserted inward to the appropriate layer, and is reoriented; the buffer surface is finished last. The completed simplified solve demonstrates that setup-conjugated local operations can make a very large four-dimensional puzzle manageable. It does not establish that the same words preserve the 32 additional moving orbits of the full model [4].

The ivan216 MPUlt and mc7d-kb projects provide software background: an accessible MPUlt source reference and a control model that distinguishes grips, twists, layers, recentering, and configurable keys. The latter is Ivan's fork of Aegyo's keybinding extension to Andrey's original MC7D. These projects provide relevant comparisons for integration and interface choices. Neither is presented here as the source of the full 35-orbit construction, and the present modifications do not imply their authors' endorsement [8, 9].

3 The retained model and its invariants

3.1 Geometric foundation and scope

The regular base polytope has Schläfli symbol $\{3, 3, 5\}$, face vector $(120, 720, 1200, 600)$, and a boundary radially homeomorphic to S^3. Its full geometric symmetry group is $W(H_4)$, of order 14,400, with 7,200 orientation-preserving elements. These are rigid symmetries of the polytope, distinct from the legal twist group and the continuous $SO(4)$ camera group. Sticker interiors are three-dimensional regions of tetrahedral facets; the rank defined below counts cap memberships rather than spatial dimension [15].

The supplement's exact arithmetic checker verifies base-polytope coordinates, incidences and explicit symmetries. Its radial argument supplies the topological identification; mod-2 homology ranks alone do not. Neither calculation reconstructs the 433 cut regions per cell or proves an exact order for the legal twist group. The retained cut and controller certificates remain separate inputs.

3.2 Stable addresses and legal actions

The construction is specific to the `600-cell-Full` profile, with cuts ± 0.968, `FixedMask 2`, and no `Simplified` flag. Changing these definitions changes the problem. Geometry is reconstructed from golden-ratio coordinates with floating-point region identification; subsequent state transitions use exact integer permutations. This is not a claim of certified exact geometric clipping. A native bridge later checks the correspondence between this enumeration and the inherited MPUlt runtime [1, 2, 7].

Let $M(P)$ be the set of turning caps containing a physical piece P, and $H(P)$ its hosting cells. The rank and sticker count are

$$r(P) = |M(P)| - 1, \qquad k(P) = |H(P)|. \tag{1}$$

A position is identified by its complete cap-membership signature $M(P)$, and a sticker slot by the pair $(M(P), h)$, where h belongs to $H(P)$. Rank is a useful descriptive attribute, but it is not a unique orbit identifier. The toolkit's O00 through O34 labels are enumeration-specific names, not original MPUlt menu labels.

For each cell c, H_c and T_c denote a legal half-turn and third-turn of its cap. There are 1,200 positive generators across 600 cells. The integer encodings are $2c + 1$ and $2c + 2$, with negative integers denoting inverses. A state array L records the home identity currently occupying each slot. For a source-to-destination permutation g, the update rule is

$$L'[g(s)] = L[s], \qquad s = 0, \ldots, 259799. \tag{2}$$

Words execute chronologically from left to right. Accordingly, $[U, V]$ means $UVU^{-1}V^{-1}$, and the cycle $(a\ b\ c)$ sends a to b, b to c, and c to a. Explicit conventions are necessary because reversing composition conventions can reverse a setup, a cycle, or a replay certificate.

3.3 Census and local orientation

The 35 moving orbits contain 176,520 pieces. The 600 fixed centers are 600 singleton position orbits; counting them literally gives 635 position orbits, while grouping centers as one geometric family gives 36 families. The complete 35-row census and controller-cost table are reproduced in Appendix A; the accompanying reference retains every algorithm card [2].

Table 2: Local orientation groups of the 35 moving orbits.

Local orientation group	Moving orbits	Practical consequence
Trivial	The remaining 24 orbits	No independent local orientation variable; this includes some two-sticker families
C_2	O00, O03, O08, O11, O22, O25, O32	Two local orientations; separate even flip parity in each orbit
C_5	O17, O28	Five orientations; separate transported-frame twist sum modulo five
D_5, order 10	O06	Rotation and reflection-type states; reflection parity is constrained
A_5, order 60	O34	Nonabelian corner orientation; a single residual orientation can be legal

Pairs O04/O05, O12/O13, O18/O19, and O26/O27 share rank and sticker count while remaining distinct move orbits. Combining them in a filter is legitimate; identifying them as the same orbit is not. Similarly, twenty corner stickers do not imply a twist variable modulo twenty. The attainable orientation group of O34 has sixty elements and noncommuting multiplication.

The invariant audit records even positional parity separately in every moving orbit. An odd permutation in one orbit cannot be canceled by an odd permutation in another. For O06, a reflection-type element of D_5 exchanges two pairs of five stickers and is therefore even as a sticker permutation. Ordinary sticker-permutation sign consequently does not detect its reflection bit. Orientation must be analyzed in the correct local group [1, 2].

These statements concern legally tracked labelled states. Assigning arbitrary identities to visually indistinguishable pieces can manufacture an apparent parity defect. Exact identities should be reconstructed from legal history, rather than guessed from the displayed colors.

3.4 Piece permutations, frames, and the ambient group

Write $\mathcal{S}$ for the 259,800 sticker slots, $\mathcal{P}$ for the physical piece positions, and $\rho : \mathcal{S} \to \mathcal{P}$ for the sticker-to-piece map. A legal sticker permutation g induces a piece permutation $\bar{g}$ because all stickers belonging to a piece travel together:

$$\rho(g(s)) = \bar{g}(\rho(s)) \qquad (s \in \mathcal{S}). \tag{3}$$

This is stronger than asking that the color array contain the right number of each color. A bijective array can still split a piece or exchange pieces between distinct orbits. The implementation checks label bijectivity, agreement of the occupant identity across every sticker of a piece, and equality of source and destination orbit IDs. These checks establish structural consistency; arbitrary arrays passing them are not thereby proved reachable. Reachability evidence is supplied by replaying legal history [11].

For an orbit o, choose a reference ordering of the stickers at each position. Its N_o pieces have attainable local frame group H_o. A transported action can then be described by a position permutation p and a frame increment $a_i \in H_o$ at each source position i. In a right-action convention, (i, h) is sent to $(p(i), ha_i)$. Applying (p, a) and then (r, b) gives

$$(p, a)(r, b) = (r \circ p, c), \qquad c_i = a_i b_{p(i)}. \tag{4}$$

The reindexing of b is essential: orientation increments belong to transported pieces, not to independent counters at fixed screen locations. Different choices of reference frames conjugate the coordinate description without changing the legal action. The full moving group embeds in a product of permutation wreath products,

$$G \;\hookrightarrow\; \prod_{o=0}^{34} (H_o^{N_o} \rtimes S_{N_o}). \tag{5}$$

This inclusion is an upper description, not an assertion that arbitrary independent permutations and orientations are legal. All 600 center slots are fixed and contribute no multiplicative state factor. Appendix A supplies every orbit's size, rank, sticker multiplicity, orientation group, and controller cost.

3.5 Abelian invariants and an upper count

Let $H_o^{\mathrm{ab}} = H_o/[H_o, H_o]$ and let $\chi_o : H_o \to H_o^{\mathrm{ab}}$ be the quotient map. Written additively in this abelian quotient, the aggregate transported orientation is

$$\Phi_o(p, a) = \sum_{i=1}^{N_o} \chi_o(a_i), \qquad \Phi_o((p, a)(r, b)) = \Phi_o(p, a) + \Phi_o(r, b). \tag{6}$$

The second identity follows from Equation (4): a permutation merely reorders the terms contributed by b. If every primitive has zero image, every legal word has zero image. This explains why the invariant is checked on generators rather than inferred from a few solved scrambles. The original transported-frame generator audit is retained evidence; the newer independent group checks verify the finite local groups and their commutator images, without claiming to rerun that entire geometric audit.

For C_2 and C_5, the abelianization is the group itself, giving the separate flip and twist-sum constraints. For D_5, with presentation $R^5 = F^2 = 1$ and $FRF = R^{-1}$, its derived subgroup is $\langle R \rangle$ and its abelianization is C_2. Thus a final residual can be rotational even when its reflection bit must vanish. For A_5, the abelianization is trivial. The word "reflection" here describes the abstract action on the five local stickers, not a claim that a legal move reflects the ambient four-dimensional space.

Ignoring all constraints first gives the labelled structural count

$$U_{\text{struct}} = \prod_o N_o! \, |H_o|^{N_o}, \qquad \log_{10} U_{\text{struct}} \simeq 598075.039945. \tag{7}$$

Within this ambient product, imposing even position parity independently in 35 orbits, seven C_2 orientation sums, two C_5 sums, and one D_5 reflection parity gives

$$|G| \le U_{\text{inv}} = \frac{U_{\text{struct}}}{2^{35}\, 2^8\, 5^2}, \qquad \log_{10} U_{\text{inv}} \simeq 598060.697715. \tag{8}$$

Each constraint has the stated index in the ambient product; imposing them does not establish their sufficiency for the actual legal subgroup. In particular, this labelled upper bound must not be substituted for the face-color lower bound in Section 4. Additional inter-orbit restrictions, if present, can only reduce the reachable group. The report's strong comparison with 4^7 requires neither equality in Equation (8) nor an exact full-group order.

4 A conservative configuration lower bound

4.1 Construction and verification obligations

The full reachable group order is not needed to establish a substantial scale gap. Consider the nine single-sticker orbits

$$\mathcal{I} = \{\text{O01}, \text{O04}, \text{O05}, \text{O07}, \text{O18}, \text{O19}, \text{O27}, \text{O31}, \text{O33}\}. \tag{9}$$

Their sizes are 3,600 for O01, 2,400 for O33, and 7,200 for each of the other seven. The audit replays each raw seed over all 259,800 labelled slots and checks that its complete effect is precisely one three-cycle in the stated orbit, with all other slots fixed. It also checks the relocation to the designated buffer positions, independently rebuilds the guarded setup search, and verifies coverage of all $N_o - 2$ nonbuffer destinations. Every orbit in $\mathcal{I}$ has one sticker per piece and trivial local orientation.

Let the relocated pure controller be $(A\ B\ C)$. For every nonbuffer X, a legal setup S fixes A and B and takes C to X. The conjugate $S^{-1}(A\ B\ C)S$ is therefore $(A\ B\ X)$. These star three-cycles generate A_{N_o}, the alternating group on that orbit. As the nine families have disjoint support and each controller is pure on the complete puzzle, their alternating-group actions can be combined independently.

The solved state has 600 distinct cell-color symbols. Within each selected orbit, every symbol occurs $m_o = N_o/600$ times: six, four, or twelve. The number of color arrangements realized by A_{N_o} equals $N_o!/(m_o!)^{600}$. To see why no factor of two is required, note that m_o is at least two. Exchanging two pieces of the same color is an odd permutation that stabilizes the coloring. Any odd representative of a color arrangement can thus be replaced by an even representative with the same visible assignment.

4.2 Bound and comparison

It follows that the full model has at least

$$B = \frac{3600!}{(6!)^{600}} \frac{2400!}{(4!)^{600}} \left[\frac{7200!}{(12!)^{600}}\right]^7 \tag{10}$$

distinct face-color states, with every unselected orbit held solved. Numerically, $\log_{10} B$ is approximately 151851.164322. This is a lower bound derived from a specific verified subgroup, not an asserted exact order for the full puzzle group and not a count of individually labelled stickers. The detailed generation and color-quotient arguments follow below.

A side-length-four seven-dimensional hypercube has $2d\,n^{d-1} = 57{,}344$ stickers. With fourteen colors and 4,096 copies of each color, the number of all balanced face-color assignments is

$$U = \frac{57344!}{(4096!)^{14}}, \qquad \log_{10} U \approx 65695.470510. \tag{11}$$

Every legal 4^7 state is among these assignments, so U is an upper bound regardless of its further permutation and orientation restrictions. Fixed-frame counting is used for both puzzles. Conservative outward rounding gives $B > 10^{151851}$ and $U < 10^{65696}$, hence $B/U > 10^{86155}$.

This argument uses comparable face-color states on both sides; it does not compare a labelled upper bound for one puzzle against a colored lower bound for another. It is also intentionally limited to a named, documented solved high-dimensional comparator. A configuration-space gap does not by itself yield a cognitive difficulty ratio or a claim that no larger puzzle has been solved. A word-metric lower bound additionally requires a specified move alphabet; a conservative version is derived below. The reproducible audit and numerical calculation accompany the report.

4.3 Why the selected stars generate independent alternating actions

Proposition 4.1 (Pure-star reachability)**.** *Suppose an orbit has $N \geq 4$ single-sticker positions and legal words realizing every pure cycle $t_X = (A\ B\ X)$, for $X \notin \{A, B\}$, while fixing every sticker outside that orbit. Then every even permutation of the orbit is realizable without net movement outside it.*

Proof. For distinct nonbuffer positions X, Y, chronological composition gives

$$t_Y^{-1} t_X = (A\ Y\ X). \tag{12}$$

Together with the original stars and their inverses, these supply every three-cycle containing A. For three distinct positions $x, y, z \neq A$,

$$(A\ x\ y)(A\ z\ x) = (x\ y\ z). \tag{13}$$

Every even permutation is a product of three-cycles. Conversely, each star is even, so the generated action is exactly A_N. Purity is preserved by multiplication and inversion, proving the support claim. The two displayed identities are also checked independently on small permutation domains in the public arithmetic supplement. □

The nine audited pure families have disjoint supports. Actions on disjoint supports commute, and an element common to two such supported subgroups must be the identity. Their product therefore embeds as a direct product, which justifies multiplying the nine color-arrangement factors in Equation (10). This independence is obtained from explicit full-support controllers, not from the fact that the pieces happen to have different orbit names.

For completeness, the color quotient can be expressed by orbit–stabilizer. The stabilizer of the solved coloring inside S_N is $(S_m)^{600}$, where $m = N/600$. Since $m \geq 2$, this stabilizer contains an odd same-color transposition. Hence its intersection with A_N has half its order, and

$$\frac{|A_N|}{|A_N \cap (S_m)^{600}|} = \frac{N!/2}{(m!)^{600}/2} = \frac{N!}{(m!)^{600}}. \tag{14}$$

The cancellation is why the lower bound counts observable cell-color arrangements rather than artificial distinctions among identical colors. A display palette with repeated RGB values may visually identify further states; the mathematical alphabet remains the 600 distinct home-cell symbols.

4.4 Word-metric consequences and the limits of short scrambles

A move-count argument becomes possible only after fixing an alphabet. Here use the lab's 600 half-turns and 600 third-turns with their inverses: $q = 1800$ distinct elementary symbols. A half-turn is its own inverse. Native multi-cap commands and macro applications are different units and are not counted as single letters in this metric.

Let Ω be the reachable fixed-frame face-color states and let D be their maximum distance from solved in this alphabet. Since $|\Omega| \geq B$, the elementary word-ball bound is

$$|\{x : d(1, x) \leq d\}| \leq \sum_{j=0}^{d} q^j = \frac{q^{d+1} - 1}{q - 1}. \tag{15}$$

Distinct words can produce the same state, so this deliberately overcounts. Exact integer comparisons in the supplement show that the right side at $d = 46647$ is smaller than B. Therefore

$$D \geq 46648 \quad \text{lab primitive turns.} \tag{16}$$

This is an existence bound on worst-case distance, not a claim that every scramble requires 46,648 moves. It also does not establish that the full puzzle has a greater diameter than a different solved puzzle. A state produced by a known 1,000-turn scramble always has its inverse word as a solution of at most 1,000 turns, irrespective of how costly a general constructive method is.

There is an even sharper distinction between a random walk and a uniform random state. The support of a 1,000-step scramble distribution μ_{1000} has at most 1800^{1000} states. If ν is uniform on Ω, total variation distance, defined by $\|\mu - \nu\|_{\mathrm{TV}} = \max_A |\mu(A) - \nu(A)|$, obeys

$$\|\mu_{1000} - \nu\|_{\mathrm{TV}} \geq 1 - \frac{1800^{1000}}{|\Omega|} \geq 1 - \frac{1800^{1000}}{B} > 1 - 10^{-148595}. \tag{17}$$

The strict numerical inequality is verified with integers; $1000 \log_{10} 1800 \simeq 3255.272505$ only explains its scale. This conclusion applies to a 1,000-step walk from the solved state and says nothing about whether its appearance is sufficiently disordered for a practical trial.

For the idealized independent draws represented by the application's generator distribution, probabilities are symmetric under inversion: each half-turn has probability 1/1200, and each signed third-turn has probability 1/2400. The induced finite walk therefore has uniform stationary measure. It is irreducible on its reachable orbit by definition of the generating set, and return words of lengths two and three make it aperiodic. Thus asymptotic convergence of the idealized walk is compatible with Equation (17); no useful mixing-time estimate follows. A fixed seeded pseudorandom implementation is deterministic and is not itself a proof of exact independent sampling. The three archived trials test deterministic legal replay and controller behavior on their recorded walks, not typical behavior under the uniform distribution [11].

5 Constructive solving across all 35 orbits

5.1 Seeds, collateral, and two execution modes

For orbit o, the reference supplies a legal seed word W_o whose target-orbit effect is a three-piece cycle. Twelve raw seeds have no collateral; the remaining twenty-three also move other orbits. A filtered image of the target cycle cannot determine which case applies. For example, the transcript gives a short word that cycles pieces in both O24 and O27; hiding rank fourteen conceals the second action [1, pp. 7–8]; [2].

Two execution modes follow. A pure controller is constructed by applying a raw seed and then correcting all its collateral through previously constructed pure controllers. The measured dependency graph is acyclic, so these corrections form a finite directed acyclic graph of legal words. Purity describes the complete net effect: intermediate primitive turns may disturb pieces that return home before the macro finishes.

The shorter production method instead retains raw seeds and solves each target orbit before the orbits that its seed disturbs. All collateral then falls within unfinished stages. This triangular method is the workload baseline used by the archived synthetic solutions. Executing only the visible three-cycle while discarding its other effects would implement a different, uncertified operation.

5.2 Star operations and guarded setups

Let R_o relocate a seed's three targets to the reference positions A, B, C, and let $S_{X,q}$ carry the reference at C to a selected destination X with attainable ordered sticker frame q, while fixing A and B. A positive star is

$$K_{o,X,q} = S_{X,q}^{-1} R_o^{-1} W_o R_o S_{X,q}. \tag{18}$$

Its target-orbit action is $(A\ B\ X)$; its inverse acts as $(A\ X\ B)$ and also counts as one star. Buffer locations are fixed during setup, but their occupants move when the star executes. This distinction prevents the mistaken assumption that a buffer permanently contains the same physical piece.

The primitive cost is

$$|K_{o,X,q}| = |W_o| + 2|R_o| + 2|S_{X,q}|. \tag{19}$$

For O33, the pure seed $[[[[H_0, T_1], H_{15}], H_{56}], H_{21}]$ has a stored length of 46 primitives. Its relocation adds ten moves, making a reference star 56 moves long; a target setup of depth d gives $56 + 2d$. Exact forward and inverse words, without ellipses, are retained for all thirty-five seeds in the reference companion [2].

Setup searches forbid primitive caps belonging to $M(A) \cup M(B)$. Their states include both the destination and its ordered sticker frame. The retained trees cover $(N_o - 2)|\Omega_o|$ oriented nonbuffer states for every orbit. For O34 this is $(120 - 2) \times 60 = 7{,}080$ states. Thus the method specifies finite lookup paths, rather than leaving setup discovery as an informal instruction.

5.3 Placement and orientation

For an incorrect nonbuffer destination X, let $Y = \mathrm{where}(X)$ be the current position of the piece whose home is X. If $Y = A$, use K_X^{-1}; if $Y = B$, use K_X; otherwise, when Y differs from A, B, and X, use $K_Y^{-1} K_X$. The latter has target-position action $(A\ Y\ X)$, inserting the required piece at X while preserving already completed nonbuffer positions. Frame choices are included in the fully specified stars.

Once every nonbuffer position is correct, an isolated swap of A and B would be odd. The separately checked even positional parity in that orbit excludes this residual. This is an orbit-specific argument; combining parity across different families would be incorrect.

Orientation is then controlled by comparing a canonical star with a star using another attainable frame:

$$P_X(q) = K_{X,q} K_X^{-1}. \tag{20}$$

This fixes positions and transfers inverse orientation changes between X and a buffer. A finite search among actual frames chooses the correction that restores X's labelled stickers. After orienting the nonbuffer positions and then the first buffer, only the final buffer can retain a defect. The generic final-buffer construction uses $[P_X(q), P_Y(r)]$ for distinct nonbuffer positions. Its eight constituent stars change only that buffer's orientation on the target orbit.

For C_2 and C_5 families, the separate flip or twist-sum invariant forces the final residual to vanish. For D_5, the residual lies in its rotational commutator subgroup; if R is a five-way rotation and F a reflection-type element, $[R^k, F] = R^{2k}$, and multiplication by two is invertible modulo five. For A_5, the retained finite commutator table covers the required residuals. The argument uses actual group multiplication rather than a scalar approximation to corner orientation [1, pp. 9–11]; [2, pp. 4–6].

5.4 Preserving completed work

The safe raw-seed stage order is supplied with the data and follows the actual collateral dependencies, not monotonically increasing rank. After every macro, the solver applies its complete permutation. After every stage, it checks that all previously completed orbits remain solved. The final acceptance condition is $L[s] = s$ for every slot s.

Reduction, blockbuilding, and orbit-first solving differ primarily in preservation costs. Forgetting all but O00, O06, and O34 gives a legitimate simplified quotient. Solving that quotient leaves a kernel problem on the additional full-puzzle pieces; it does not justify treating omitted regions as permanently bandaged blocks. Graph-shell blockbuilding can use pure controllers to preserve completed pieces at macro boundaries, but does not promise that they remain motionless throughout each expanded word. Orbit-first solving with cell-local destination batching retains the tested dependency order while reducing navigation overhead. The retained evidence supports it as the practical baseline [1, pp. 12–13].

5.5 Guarded setup search as a finite graph problem

The setup table is a graph certificate. A node records an ordered sticker frame $f = (s_1, \ldots, s_{k_o})$ at a nonbuffer destination. Applying an allowed primitive sends it to $g(f) = (g(s_1), \ldots, g(s_{k_o}))$. The allowed positive alphabet excludes every cap in $M(A) \cup M(B)$, so both buffer locations and their sticker frames remain pointwise fixed. The source frame at C is the root. A breadth-first traversal stores one parent, one move, and one depth per discovered frame:

> **Guarded-frame construction.** Initialize the queue with the ordered frame at C, with parent -1 and depth zero. Remove the next frame f. For each allowed positive generator g, compute $g(f)$; if it has not been seen, record f as its parent, store g, set its depth to $\mathrm{depth}(f) + 1$, and append it to the queue. Continue until the queue is exhausted. Accept coverage only if the distinct destination/frame count equals $(N_o - 2)|H_o|$ and the recorded transitions and guards are valid.

An inverse setup is reconstructed by reversing its parent path and inverting every move. Breadth-first depth is minimal within this particular guarded positive alphabet. It is not a shortest solution among arbitrary native moves, all inverse generators, alternative buffer choices, or setups that temporarily move the buffers. In particular, the word "shortest" should not be promoted from a graph-search property into a claim of globally optimal algorithms.

If $V_o = (N_o - 2)|H_o|$ and b_o is the number of allowed generators, traversal takes at most $O(V_o b_o k_o)$ elementary frame-coordinate transports when a move lookup is constant-time. Parent storage is $O(V_o)$; explicitly storing every ordered frame uses $O(V_o k_o)$. The implementation's sparse lookups add their own search costs. These expressions explain why the corner orbit, although containing only 120 pieces, still needs 7,080 oriented setup states. A table of destination IDs alone would discard the orientation information needed for insertion.

Adding legal inverse edges cannot increase minimum unit-cost setup length, although it can increase graph-construction work. Unequal nonnegative costs require weighted search. Likewise, conjugating legal moves by a geometric symmetry does not automatically preserve the chosen 1,800-symbol metric. A symmetry reduction must preserve the actual graph, edge costs, goals and protected frames, and retain enough information to lift a path back to the labelled puzzle [15].

5.6 Stage correctness and the role of collateral

Let a directed edge $o \to j$ mean that the raw seed for o has nontrivial net support on orbit $j \neq o$. Since every legal setup preserves each move orbit as a set, conjugation changes positions within these support orbits but cannot create support on a previously untouched orbit. This observation connects the measured collateral graph to the solving order.

Proposition 5.1 (Triangular stage invariant)**.** *Assume each target orbit has the certified setup/frame coverage and orientation corrections described above, and the raw-seed collateral graph is acyclic. If targets are processed in an order placing every o before every j with $o \to j$, then every completed orbit stays solved at subsequent macro boundaries.*

Proof. Suppose the first t target orbits are solved. The next raw seed fixes those entire orbits because all of its collateral targets occur later. Every conjugate of that seed fixes them as well, since the conjugating setup maps each such orbit to itself. The target placement step inserts a required home piece without moving completed nonbuffer targets; the final two-buffer swap is excluded by even positional parity. Orientation transfers fix positions and correct the nonbuffers, then buffer A, then any admissible residual at B. Thus the next orbit is solved and the previously completed ones remain solved. Induction establishes the claim. $\square$

This is a constructive argument under explicit certificate hypotheses. The published compact trials exercise all 35 stages; the new arithmetic audit separately checks that the retained graph has every collateral target later in the stated order. Neither check should be confused with a theorem-prover verification of geometric clipping or every oriented transition. The full order is printed in Appendix A so that the preservation claim can be compared with the actual data.

The proof is about *macro-boundary* preservation. During S^{-1}, W_o, or S, many previously completed pieces can move and return. A rule that bans any intermediate motion is stronger and would exclude legal witnessed macros used here. The workbench's protected-orbit check consequently tests the complete net support; it is not a physical bandaging mechanism.

5.7 Nonabelian orientation and finite residual correction

For D_5, a residual with zero reflection bit lies in $\langle R \rangle$. Since

$$[R^k, F] = R^{2k}, \qquad 2^{-1} \equiv 3 \pmod 5, \tag{21}$$

any desired rotational correction R^a is obtained by choosing $k \equiv 3a \pmod 5$. The twenty-sticker corner family requires a different treatment. Its group A_5 has one identity, fifteen elements of order two, twenty of order three, and twenty-four of order five. Noncommuting elements cannot be represented by a single additive twist counter.

Perfectness of A_5 means that its elements are generated by commutators; that statement alone does not imply that one commutator suffices. The finite calculation used here checks the stronger fact: the set $\{[q, r] : q, r \in A_5\}$ contains all 60 elements. Searching at most 3,600 ordered pairs therefore finds a single local commutator for any required residual. Expanding each transfer P_X into two stars makes the final-buffer correction eight stars. The public extended checker independently recomputes the D_5 and A_5 commutator images on explicit finite permutations.

The implementation uses a three-star word for buffer A's correction. A target-orbit identity relating this word to another construction is insufficient to replace its full-state effect: the two words can differ on unfinished collateral orbits. The engine always composes the actual stored legal recipe over all slots. This is a concrete example of why algebraic simplification must be checked in the full action, not merely in a quotient showing the target orbit [11, 2].

5.8 Sparse execution and certificate complexity

For a net permutation with moved support m, store parallel integer arrays s, d with $s_i \mapsto d_i$. Chronological composition of two supported operations is most safely defined by their full action before trimming the fixed points. If A is a destination-to-source workspace initialized to identity, the update $A[d] \leftarrow A[s]$ yields the final inverse-map representation; extracting changed destinations and their occupants recovers the source-to-destination pairs. The right-hand side must be gathered before the write, since source and destination supports overlap in cycles.

Conjugating a support by a setup does not require replaying a long word independently on every label. Under the report's convention, the support of $S^{-1}WS$ is obtained by transporting both endpoints of W through S. The implementation first replays and certifies the complete seed, then checks the transported target frames and the buffer guard for each requested star. Cached maps retain the same full collateral support as their finite legal witnesses.

The application bounds its least-recently-used star cache at 96 entries. An entry uses approximately $8m$ bytes for two 32-bit index arrays before object overhead; it does not store a multi-million-move expansion. Applying a sparse map costs $O(m)$ assignments, but refreshing exact piece progress and computing the complete state hash still scans the full sticker array. Thus the end-to-end application cost is not simply $O(m)$. The archived high-throughput verifier has a different execution path and should not be timed as though it were the interactive application's commit pipeline.

The primitive length remains a witness cost even when the net effect is cheap to apply. A compact history with roughly 374,000 stars represents roughly 353 million primitives because the interpreter executes verified net permutations. This is semantic compression of a retained legal sequence, not a claim that the same number of elementary animations took only seconds. It also explains why an interactive importer can legitimately bound expanded work while a separate research verifier accepts the compact synthetic histories.

6 Workload, replay, and the meaning of a solve

6.1 Recorded computational results

The three synthetic full-model trials begin from 1,000-move scrambles with random seeds 600, 601, and 602. Their compact histories record every star and preserve the full collateral action. The archived totals are:

Table 3: Archived synthetic solutions. Historical solving times exclude preparation and Windows graphics.

Synthetic trial	Stars	Represented solution primitives	Historical headless solving time
Seed 600	374,178	352,698,206	8.08 s
Seed 601	373,948	352,080,470	7.86 s
Seed 602	374,428	352,839,050	7.24 s

The timing column belongs to the original container experiment and excludes preparation and Windows graphics. It is retained as historical evidence, not relabelled as a current hardware benchmark. The public companion contains the three synthetic histories and a standalone verifier. Its publication audit regenerates their scrambles and replays the compact full-state effects; the audit's separate results identify exactly what was rerun.

A successful compact replay establishes that the recorded legal macro history restores the complete labelled model. Checking a saved color field alone would be weaker. Conversely, checking a huge native-format archive's syntax, counters, nesting, and checksum would not constitute primitive-by-primitive geometric replay. The original transcript explicitly distinguishes these evidence levels [1, pp. 25–30].

The historical monolithic expansions are unsuitable as ordinary native-session files. Individual decompressed logs are approximately 2.9 GB, and the old 32-bit sequence allocator can require still more memory during growth. They are not included in the public companion. Compact certificates preserve the legal sequence without requiring such files to be loaded into the GUI. The 0.2.4 bounded log importer should likewise not be described as an importer for arbitrary multi-gigabyte histories.

The reference also gives a conservative method-specific bound of 405,945 stars and 385,797,152 represented primitives. The independent audit recomputes this bound from the retained orbit sizes, orientation groups, and maximum setup depths. It bounds this construction, not the shortest possible solution [2].

6.2 Human workload and assistance

A star is an algebraic unit, not a solved piece or an elementary hand action. Placement may require two stars, and an orientation transfer typically requires two. In the seed-600 trace, placement accounts for 345,833 stars, nonbuffer orientation for 28,302, first-buffer orientation for 27, and final-buffer commutators for 16. The mean represented cost is approximately 943 primitive turns per star [1, pp. 27–31]; [2].

For a scenario with t seconds of recognition, target/frame selection, checking, and application per star, the active-time model is

$$T_{\text{hours}} = \frac{374178\,t}{3600}. \tag{22}$$

At 15–30 seconds per star this gives approximately 1,559–3,118 active hours, before learning, development, breaks, or recovery. These are throughput scenarios, not observations of human performance. At one primitive turn per second, the represented seed-600 solution would require about 97,972 hours of uninterrupted execution. Neither estimate is a lower bound on the puzzle's optimal solution length.

The distinction between manual, macro-assisted, target-assisted, and automated solving must therefore be recorded explicitly. Assistance that chooses targets and frames changes the nature of an attempt even when every selected macro remains legal. Primitive counts, star counts, transactions, assistance labels, and active time should be retained separately. The framework demonstrates a complete computational route, while a human attempt remains an additional empirical undertaking.

6.3 A reproducible bound for the stated construction

Let a_o indicate a nontrivial local orientation group, and let b_o indicate one of the nonabelian groups D_5, A_5. At most two stars insert each of the $N_o - 2$ nonbuffer positions. When orientation is nontrivial, at most two stars correct each nonbuffer, three correct the first buffer, and eight handle a possible nonabelian final-buffer residual. Hence the method-specific per-orbit bound is

$$Q_o = 2(N_o - 2) + a_o[2(N_o - 2) + 3 + 8b_o]. \tag{23}$$

If $d_o^{\max}$ is the retained maximum setup depth, a conservative primitive bound uses

$$C_o^{\max} = |W_o| + 2|R_o| + 2d_o^{\max}, \qquad Q = \sum_o Q_o = 405945, \quad \Lambda \leq \sum_o Q_o C_o^{\max} = 385797152. \tag{24}$$

This bound includes the chosen worst-case setup cost for every star and is consequently looser than summing the actual visited depths. The public arithmetic checker recomputes all 35 terms directly from the census and execution trees. Table 6 makes the largest seed and setup costs visible instead of hiding them in one aggregate count.

There are three separate complexity questions: finding the controller bank, selecting and applying a solution using that bank, and expanding or animating its witness. The finite setup trees address the first; the constructive placement/orientation procedure addresses the second; the primitive counts address the third. The costs should not be substituted for one another. Equation (16) is a lower bound on a specified worst-case word metric, while Equation (24) is an upper bound for a particular solving method under its certificate assumptions. Neither is an estimate of an optimal human algorithm.

Fresh replay of seeds 600, 601, and 602 checked complete labelled identity at completion and preservation of all 35 stages. The recorded replay-only times in that publication audit were 13.19, 29.28, and 32.25 seconds; preparation and total process times are recorded separately inside the companion. These runs were validation observations, not controlled performance comparisons between scrambles. They are therefore not used to infer a statistical average solver speed or a native rendering rate.

7 From construction to a dependable workbench

7.1 State, certificates, and persistence

The transcript's architectural decision is to make the 259,800-entry labelled array authoritative and treat rendering as a derived view. In 32-bit integers this array occupies 1,039,200 bytes, about 0.99 MiB. Geometry and projection, rather than the logical state alone, account for much of the resource burden.

A macro certificate identifies the model, legal recipe, full permutation digest, support by orbit, primitive/star counts, and before/after state hashes. These digests identify data and state transitions; they are not digital signatures proving authorship. The engine rechecks the preview revision and protected-orbit intersection at commit. It then applies the complete net effect as one durable transaction, retaining the legal witness for independent expansion or replay.

The journal preserves parent-linked history and checkpoints rather than deleting alternative branches. Reset and validated imports create recovery paths; undo and redo verify expected state transitions. A coherent native snapshot combines state, status, colors, and styles, avoiding a display assembled from different revisions. Filters and motion sampling change visibility only. Hidden pieces continue to receive every mechanical update and remain included in progress and solved-state checks [1, pp. 35–39]; [7].

7.2 Integration failures as evidence

The original transcript documents a real verification gap: source checks, layout arithmetic, and software-rendered browser tests did not establish that the native WinForms/DirectX path worked on Windows. A splitter initialization defect and a launcher assumption about matching parent/interpreter process IDs survived those earlier checks. The later local work also addressed renderer lifecycle and resize recovery, color/style synchronization, hidden-piece picking, and frame visibility [1, pp. 47–52]; [7].

The frame defect illustrates why a display policy must apply at every draw entry point. Guarding only the normal lifecycle path allowed inherited direct or animated redraws to bypass frame hiding. The repaired scene boundary applies the same policy to those redraws; a visible checkbox and View-menu item share a persisted setting. Visibility-aware picking rejects hidden pieces even when a frame is intentionally shown. These changes make filtering a reliable instrument for inspection while preserving the complete puzzle.

7.3 Subsequent native validation

The final 0.2.4 sources passed 68 comprehensive native checks and four fresh-process reopen checks using the actual MPUlt runtime, official installed Managed DirectX assemblies, native controls, viewport messages, and render-target pixels. Coverage includes twisting, 3D/4D rotation, colors, scrambling and undo/redo, filters and picking, keys, checkpoints, macros, buffer/insertion tools, reports, reset, log exchange, and controlled device-loss recovery. Three full-detail poses and a filtered selection view matched the comparison rendering path exactly [7].

Backend mechanics, independent generator maps, crash recovery, frame preferences, process lifecycle, frozen packaging, and a real isolated install/uninstall were checked separately. Normal executable startup preserves an existing session and does not run the developer WinForms fixture. These results close specific earlier native integration gaps. They do not retroactively turn the transcript's original container timings into native measurements, and they do not establish that hundreds of millions of primitives were individually animated to completion.

7.4 Component boundaries and the native equivalence contract

The executable is a composition of a 64-bit Python state engine and a 32-bit .NET native host. This split permits the inherited MPUlt/Managed DirectX viewport to retain its compatible process architecture while the authoritative state and research tools use a separate runtime. Packaging freezes the engine and launcher; the user does not need an external interpreter or compiler. Build-time layout fixtures remain developer checks and are excluded from ordinary startup [11, 7].

The bridge is more than a color-index conversion. It first identifies all 600 cell normals and verifies the expected partition of 433 stickers per cell, the 300 antipodal axes, the cut values, and the fixed middle layer. Each native sticker is assigned its complete cap-membership signature and hosting cell. Matching these pairs to the lab enumeration constructs a bijection $b : \mathcal{S}_{\text{native}} \to \mathcal{S}_{\text{lab}}$.

For every matched native generator g_N and lab generator g_L, the required commuting relation is

$$b \circ g_N = g_L \circ b. \tag{25}$$

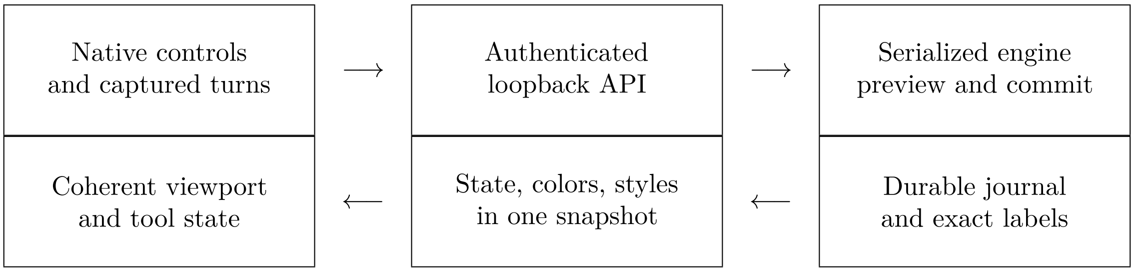

Figure 1: Logical request and response flow. The diagram shows data ownership and publication boundaries, not separate processes for every box.

The check compares complete source-to-destination maps, not a sample of vertices or a solved screenshot. Once it holds for every generator, induction extends it to every translated finite word. All 1,200 lab generators must be accounted for before normal native input is enabled. Numerical normal matching is still part of the enumeration correspondence, so this is a certificate for the retained runtime/model pair rather than a general theorem about every possible MPUlt build.

The bridge also enumerates the twelve local rotations of each tetrahedral cap from the lab generators, translating the native axis/twist/angle/layer tokens into finite words. The two outer antipodal caps are disjoint, permitting the combined outer-layer mask to be translated as a concatenation. A native turn request carries the expected pre-state hash; stale native input is rejected rather than silently applied to a newer engine state. The host treats inherited turn history as a temporary input record, then replaces its displayed fields from the committed engine snapshot.

7.5 Preview, validation, and the durable commit boundary

A preview is a prepared state transition, not a state mutation. Its data include the normalized legal recipe, complete supported map, expected next labels, full pre/post hashes, current journal head, process-local revision, and an unpredictable one-use token. The certificate digest binds its core model/recipe/map/count/state fields. Annotations and assistance labels provide interpretation; a checksum is not an authorship signature or a proof that the person naming an algorithm invented it.

Let h be the current journal head, r its in-memory revision, and $\mathcal{P}_{\text{guard}}$ the currently protected orbits. Abstractly, commit accepts only when

$$\begin{aligned} \text{token} &= \text{pending token}, \\ (h, r, \text{Hash}(L)) &= (h_p, r_p, \text{Hash}(L_p)), \\ \text{orbits}(\text{supp}(g)) \cap \mathcal{P}_{\text{guard}} &= \varnothing. \end{aligned} \tag{26}$$

The head distinguishes history branches; the revision detects an intervening undo/redo or checkout even if a previous label state is revisited; the hash binds the exact stickers. Protected-orbit preferences are re-read at commit. View-only changes need not invalidate a mathematically valid preview, but newly protected collateral must still reject it.

> **Commit protocol.** Under the session lock, validate the pending token and Equation (26). Construct the next exact state. In one SQLite transaction, append the event with its parent and pre/post hashes, write the new head and preferences, and write an automatic snapshot when required. Only after that transaction commits, publish the new in-memory head and labels, invalidate the preview, clear obsolete redo selection, and increment the revision. Return one coherent status snapshot.

This ordering defines the durable boundary. A failure before database commit must not appear as a completed operation. A failure while preparing the response after commit must not revive the old preview or implicitly reverse the committed event. The scramble path specifically preserves an earlier pending operation on cancellation only if both the head and state hash are still unchanged. It otherwise respects the already durable result. This distinction addresses a common source of apparently inconsistent scramble colors: logical commit and view refresh are related stages, not one indivisible graphics call.

The HTTP server uses a session lock around state work and a single-worker executor for long analysis operations. This serializes mechanics while allowing progress polling and cancellation through the service. It does not imply that a single SQLite connection can safely accept arbitrary concurrent writes without that application lock. Cancellation is checked during lengthy word/map work and detached import validation; a durable commit is recovered through journal operations, not by pretending that it never happened.

7.6 Branch history, checkpoints, and restart recovery

Each event has a parent pointer and records a finite recipe, before/after hashes, witness counts, assistance category, and note. Undo applies the inverse recipe and verifies the parent's expected hash. Redo checks both the selected child's parent and pre-state before replaying it. If history has multiple children, branch choice is explicit; creating a new child does not delete the abandoned continuation.

Recovery follows the chosen head backward to an available ancestor snapshot, validates that snapshot's labels, and replays the remaining chain while checking every pre/post hash. Automatic snapshots are inserted at depths divisible by fifty; named snapshots add convenient restart points. Reset first creates a recovery snapshot, then moves the selected head to the immutable solved root while retaining view preferences. Checkpoint restore and history checkout are different operations: the former restores saved preferences with the snapshot, whereas branch checkout reconstructs the requested historical state under the current workflow's rules [11].

The journal requests SQLite WAL mode with `synchronous=FULL`. SQLite documents that FULL synchronizes the WAL at each transaction commit. Its database-level WAL checkpoint, which transfers committed pages back into the main database, is distinct from a C600 puzzle checkpoint containing labels and preferences. A live WAL file is part of database state, so copying only the main file is not a general backup procedure. These are database guarantees and operational requirements; hardware durability still depends on the filesystem and storage stack [12, 13]. The recorded hard-exit tests check post-commit recovery and rollback of an uncommitted head update. They do not constitute physical power-loss testing.

7.7 Exact filtering, color transport, and hidden-piece interaction

Filters act on physical piece positions, then lift their result through ρ to all associated stickers. An identity predicate follows the occupant as it moves; a position predicate remains attached to a location. Thus `piece(p)` and `position(p)` have deliberately different meanings, as do home-cell membership and current-cell membership. The parser evaluates a bounded Boolean grammar with recursive descent; it does not execute user expressions as Python code.

If predicates $F_1, \dots, F_t$ have styles $c_1, \dots, c_t$, first-match semantics can be expressed as

$$j(p) = \min\{i : F_i(p) = 1\}, \qquad v(p) = \begin{cases} c_{j(p)}, & \text{if a match exists,} \\ 0, & \text{otherwise,} \end{cases} \qquad v_{\text{sticker}}(s) = v(\rho(s)). \tag{27}$$

An explicit context option may subsequently show buffers, selection, or preview support outside this result. Such context must not be mistaken for a failed filter. Ordinary committed scrambling avoids exposing its nearly whole-puzzle support as a transient pinned preview. The framework is controlled separately: hiding the 600-cell frame must not erase the clipping planes used to render visible stickers correctly.

Mechanical updates are independent of Equation (27). For a lab slot s, its home-cell color symbol is $\lfloor L[s]/433 \rfloor$. If f maps native cell IDs to lab cell IDs, the native color for slot n is

$$C_N[n] = f^{-1}\left(\left\lfloor \frac{L[b(n)]}{433} \right\rfloor \right). \tag{28}$$

The engine computes status, this color array, and mapped styles under one lock in one native snapshot. Its payload contains 519,600 raw color bytes and 259,800 style bytes before Base64 and JSON overhead. The host validates the profile identity, exact lengths, color range, and style range before applying the result. Three independently fetched arrays could instead describe different committed revisions.

Hidden geometry must also be unpickable. The native adapter removes hidden triangles from the picking path and tests only visible, nonclipped sticker meshes. It gates the viewport's actual release message before the inherited click handler can reuse a previous axis after an empty click. After a sampled motion frame, it first restores a current projection of every visible mesh; otherwise picking would use stale projected coordinates. Drag release, click dispatch, and nested messages during animated turns have separate guards. These measures explain why changing only a sticker's apparent transparency did not fully solve the original filter problem.

7.8 Macro tools, insertion analysis, reports, and keybindings

Macros are declarative words or stars with finite witnesses. The composition utility constructs inverse words, concatenations, commutators, and conjugates, then routes the result through the same preview and protected-support checks. In the utility API, `conjugate(a,b)` builds aba^{-1}; the star construction uses its separately stated $S^{-1}WS$ convention. These are compatible choices of parameter order, but silently interchanging them reverses setup meaning.

The buffer analyser reports the fixed reference pair, the requested destination, every attainable frame node there, its setup word and depth, and the resulting primitive cost. It does not claim an implemented arbitrary-buffer relocation search. The insertion planner uses the inverse piece-location array to find the required home piece, finishes displaced positions before orientation-only targets, and rejects an isolated odd buffer swap or an unattainable frame rather than inventing a parity patch.

Progress distinguishes correct position, wrong orientation at a correct position, exact labelled completion, and apparent color completion. The session report walks the selected head's ancestry, separating recorded scramble primitives, solution primitives, star count, transaction count, and assisted transactions. The explicit timer includes time spent thinking while it is running; it is not an inferred measure of keyboard or mouse activity. Checkpoints, key preferences, macro libraries, and saved identity/position sets are workflow state, not alterations to the retained puzzle geometry. Native key remapping must reach the same validated command path as its corresponding control, which is why the regression suite tests both the state result and the actual input dispatch.

7.9 Bounded interchange and process ownership

The interactive C600 log is a proof-oriented record rooted at labelled identity. Import first validates the entire structure and resource budget, normalizes every legal recipe, and replays all events into detached labels with exact pre/post hashes. Only a successful detached plan may replace the active journal state, with a recovery checkpoint. Imported preferences do not overwrite the current view settings; the current camera, when explicitly captured by the host, is validated separately. Original MPUlt v1 interchange passes through the independently verified native token translation.

The interactive limits are 16 MiB of file data, 24 MiB of expanded JSON, 10,000 transactions, and 2,000,000 represented primitive moves. They are deliberate service limits, not limits of the mathematical puzzle. Bounded decompression prevents a small compressed attachment from expanding without limit. The much larger research-certificate histories are verified by the separate companion tool, not imported as ordinary application logs. Saving and exporting through the native file workflow also preserve an existing destination when validation fails [11, 7].

Engine readiness is tied to a fresh launch identity and an authenticated loopback health response. A forwarding executable and its interpreter may have different process IDs, so PID equality alone is not a valid readiness criterion. Shutdown first requests termination of the owned engine and then closes the inherited parent pipe; EOF provides cleanup when a parent exits unexpectedly. Launch metadata and diagnostic encoding are explicitly UTF-8. Loopback binding, exact host/origin checks and launch tokens address this local control boundary; they do not turn the service into a remotely deployable multi-user server.

7.10 Verification layers and limits of assurance

The tests form several evidence layers rather than one undifferentiated pass count. Packed-map comparison uses a separate retained constructor but shares model geometry, so agreement cannot rule out a common clipping error. The actual native bridge compares a different enumeration and runtime action. Pixel comparison checks rendering equivalence for specified poses, while full-label comparison checks mechanical state independently of visibility. Restart and device-loss tests target failure recovery, and frozen-package tests target dependency and resource assumptions.

For camera-only, filtering, frame visibility, and layout operations, the central metamorphic condition is $\mathrm{Hash}(L_{\mathrm{after}}) = \mathrm{Hash}(L_{\mathrm{before}})$. For a word w, replaying ww^{-1} must restore every label; for primitives, H_c^2 and T_c^3 must act as identity. For rejected imports, protected moves, and stale previews, the selected head and labels must remain unchanged. These properties test semantic contracts rather than visual plausibility. Appendix B maps concrete components and regression files to the properties they establish.

The recorded 68 native checks and four reopen checks remain evidence for the frozen 0.2.4 application. This expanded report adds mathematical arithmetic and source-level analysis; it does not invent a new native benchmark run or silently broaden those test claims. A test suite reduces identified failure modes, but cannot prove that every future driver, malformed external environment, or possible sequence of GUI events is crash-free.

8 Full-detail rotation and hardware requirements

8.1 Measured behavior

The final Intel HD Graphics 620 measurements distinguish exact filtered motion, adaptive dense-scene motion, and full-detail redraw. At a 931×604 viewport, optimized full-detail frames averaged 453.45 ms, compared with 626.57 ms through the comparison path, a 27.63% reduction in elapsed drawing time. The full-detail 95th percentile was 533.82 ms. The reciprocal of that mean is approximately 2.2 FPS; the eighteen fixed-pose redraw measurements are not a sustained rotation benchmark [7].

The approximately 58 FPS result is an adaptive-motion result; it is not full-detail rotation at 58 FPS. The complete mechanical state remains active in both modes. This distinction is necessary to evaluate the requested full-detail target rather than a visually reduced substitute.

Table 4: Measured interaction and redraw performance on the tested Intel HD Graphics 620 system.

Workload on the tested system	Measured result
Exact active-orbit / work-cell motion	56.63 / 58.38 FPS
Dense-scene motion with 1,500 temporarily drawn meshes	57.84 FPS
Restore all 259,800 meshes	429.01 ms
Restore including the quiet delay after motion	630.28 ms
Optimized full-detail draw, mean / 95th percentile	453.45 / 533.82 ms

8.2 Minimum configuration: what is established

The qualification target is continuous rotation at an actual 1920×1080 **render target and at least 30 FPS, with every one of the 259,800 stickers submitted on every frame and motion detail reduction disabled**. The frame budget is 33.33 ms. No tested GPU configuration in the retained evidence satisfies that target, so a minimum GPU model cannot yet be certified. Intel HD Graphics 620 is a validated compatibility baseline, but it is not a qualifying full-detail performance configuration.

The current rendering path retains CPU-side projection and triangle/line packing. During the eighteen optimized full-detail frames, total native-process CPU consumption averaged approximately 329.86 ms per frame. Process CPU counters are not a clean separation of serialized CPU work from GPU time, but they corroborate the source-level finding that this is not a GPU-only workload. Naming a discrete GPU from specification-sheet throughput would therefore be unsupported. At the smaller tested viewport, reaching the 30 FPS budget already requires about 13.60 times less elapsed time per complete frame. Moving to 1080p also increases pixel area by about 3.69 times; frame time should not be assumed to scale by that factor because geometry and CPU costs remain.

Table 5: Established compatibility and the limits of the full-detail hardware evidence.

Requirement	Evidence-backed status
Demonstrated native configuration	Windows 10 x64, x86 native host, installed .NET Framework 4.8, and official Managed DirectX 1.1 assemblies with a working Direct3D9 hardware driver
GPU demonstrated to render the complete model	Intel HD Graphics 620, on the tested Windows system
GPU demonstrated to sustain full-detail 1080p/30 FPS	None established
Minimum VRAM guaranteeing the target	Not established; capacity alone does not certify throughput
Minimum purchase specification	No defensible GPU-only minimum until the complete CPU/GPU/driver configuration is measured

The legacy DirectX components remain an external dependency; Microsoft documents the June 2010 end-user runtime distribution [10]. A newer DirectX feature level or a larger VRAM figure alone does not demonstrate that the required legacy runtime is installed or that the CPU-bound portions meet the frame budget.

8.3 Qualification protocol

A qualifying hardware test should report the exact CPU, GPU, driver, operating system, runtime, source/build hashes, render-target dimensions, and rendering settings. It should use the same full model and complete sticker set, disable level-of-detail substitution, retain the same clipping and frame policy, and include continuous 3D and 4D rotations over fixed reproducible camera paths. A warmed run of at least sixty seconds should report frame-interval percentiles, sustained FPS, CPU use, memory use, and submitted mesh counts. A conservative pass requires sustained mean FPS of at least 30 and a 95th-percentile frame interval no greater than 33.33 ms, without missing geometry or state divergence.

Cold construction, full-detail restoration, picking latency, and stationary redraws should be measured separately. All labels must remain identical across the camera-only workload, and sampled full-detail images must agree with the established rendering reference. Until such a configuration passes, the practical tested option remains exact filtered work or adaptive motion. The open optimization problem is to reduce the complete rendering pipeline's cost, potentially including batching or GPU-side projection, and then establish a hardware minimum by measurement.

8.4 Camera mathematics and display-state separation

A useful mathematical description of four-dimensional camera orientation is $R \in SO(4)$, acting on a geometric point $v \in \mathbb{R}^4$ before projection. There are six coordinate planes. A rotation in plane (i, j) replaces the identity's corresponding block with

$$R_{ij}(\theta)\big|_{ij} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad R_{ij}^{\mathsf{T}} R_{ij} = I, \quad \det R_{ij} = 1. \tag{29}$$

The three planes within (x, y, z) describe ordinary 3D orientation changes; the three mixing w with a spatial coordinate provide the additional 4D rotations. Rotations sharing a coordinate generally do not commute. Consequently, camera history cannot in general be reduced to independent accumulated scalar angles without fixing a composition convention.

The retained viewport supplies the actual projection, clipping, shrink parameters, lighting, and picking caches. This report does not replace its projection with an approximate formula. Instead, the display dependency is written abstractly as

$$I = \mathcal{R}\big(\Pi(RV; \eta),\, C(L),\, F(L, \theta_f),\, \theta_d\big), \tag{30}$$

where V is immutable geometry, η holds projection settings, F is filtering, and θ_d includes frame and motion-detail choices. Changing $R, \eta, \theta_f, \theta_d$ does not itself apply a mechanical permutation to L. Camera validation in the current source checks finite entries, a 4-by-4 matrix's orthogonality within tolerance, and scalar bounds; it does not separately certify positive determinant. Thus Equation (29) characterizes intended rotation operations, not a claim that every structurally accepted camera matrix has been proven to belong to $SO(4)$.

8.5 Rendering lifecycle, detail restoration, and buffer reuse

The repaired lifecycle renders dirty work at most once per 16 ms UI timer tick instead of repeatedly invoking an idle renderer. It requires a live, visible, nonminimized control with positive dimensions. During camera motion, more than 6,000 visible meshes triggers deterministic sampling down to 1,500; a 180 ms quiet interval requests complete detail restoration. These are scheduling parameters, not a guarantee that a frame finishes within the nominal timer period.

The temporary render subset preserves original sticker identities and installs only the renderer-facing arrays required for that synchronous draw. It restores the complete arrays on exit, including exception paths. The scene adapter applies frame policy at the inherited scene's draw boundary so direct and animated redraws cannot bypass it. Device reset occurs with complete native data before entering a compact drawing scope; presentation occurs after that scope has been restored. This order prevents a graphics reset from encountering arrays intended only for a temporary draw.

On recognized device-lost or device-not-reset exceptions, the lifecycle clears the inherited recursion latch, marks the device unready, requests resize/rebuild, and retries after 250 ms. A stale recursion latch could otherwise suppress all later frames even after a successful reset. Other exceptions are not indiscriminately swallowed. The engineering aim is controlled recovery from known transient failures, rather than concealing arbitrary invalid state behind a blank viewport.

The full-detail optimization is deliberately narrow. It validates the pinned MPUlt assembly and the render/upload instruction hashes, creates an in-memory adapter, and reuses the managed staging array associated with an existing DirectX vertex buffer. The expected vertex stride is 28 bytes. Native projection, vertex packing, colors, order, and pick caches remain in the retained path; runtime files are not rewritten. If the verified optimization cannot be installed, the adapter can retain the original render path while keeping the same visibility/frame policy. This fallback is part of compatibility handling, not evidence that an unknown binary has passed native equivalence tests.

For a buffer containing n_v such vertices, one reusable staging array requires roughly $28n_v$ payload bytes. Reuse removes repeated allocation and marshaling work around that payload, but it does not eliminate the cost of computing projected vertices or building triangle and line data. That distinction is consistent with the measured improvement and the remaining full-detail limitation.

8.6 A pipeline model for further optimization

The relevant performance quantity is complete frame latency, not GPU specification-sheet throughput. A useful idealized model separates a fraction f of the current cost that a proposed optimization cannot accelerate from a remaining fraction accelerated by r:

$$\text{Speedup}(r) = \frac{1}{f + (1-f)/r}, \qquad \lim_{r\to\infty} \text{Speedup}(r) = \frac{1}{f}. \tag{31}$$

This is the fixed-workload bottleneck reasoning associated with Amdahl's analysis [14]. Applied only as a diagnostic model to a 453.45 ms baseline and a 33.33 ms target, even unlimited acceleration of the chosen part requires the unaccelerated fraction to be below approximately 7.35 percent. No value of f is estimated here from aggregate CPU counters: overlapping CPU/GPU work and process-wide CPU time do not provide the needed serial critical-path measurement.

The next useful experiments should therefore instrument projection, triangle/line packing, staging/upload, draw submission, presentation, and waiting separately. They should retain the exact full-detail workload and report both elapsed time and CPU/GPU measurements with their synchronization method. Candidate changes include batching fewer submission calls, reducing repeated packing, or moving mathematically equivalent projection work to the GPU. The theory supplement further describes cell instancing, complete permutation caches and incremental progress updates, each with explicit equivalence obligations [15]. Rendering changes require renewed image equivalence, picking, clipping, and camera tests; these remain research directions rather than features claimed for 0.2.4.

Hardware qualification should follow such measurements. A nominally faster GPU cannot remedy a long unaccelerated stage by itself. Conversely, a CPU-side optimization that shortens one stage does not establish the performance of an untested GPU/driver combination. The 1080p/30 FPS requirement remains a complete-configuration acceptance test, with every sticker retained and the previously stated percentile and state-invariance conditions.

9 Limitations and research directions

The retained cut profile and enumerations define the scope of the mathematics. The lower bound in Section 4 is supported by a finite model/controller audit; it does not rely on random scramble success as a proof of group reachability. The all-orbit method has explicit certificates and finite setup coverage, but the report does not claim a separately machine-checked formal proof in a theorem prover or a published exact group order.

The present seed bank emphasizes correctness over economical primitive words. In the seed-600 workload, O15, O29, and O02 account for roughly 40.3% of represented primitive moves while accounting for only about 3.8% of stars. Shorter witnesses for these stages, paired insertions, and separately certified alternative buffer pairs are concrete research targets. Replacing a seed requires rechecking its complete effect, orientation handling, dependency order, and full-state replay; a shorter word is not automatically a safe replacement [1, p. 31]; [2].

Empirical human-solving work should define its assistance policy before an attempt begins. Useful outcomes include measured recognition and target-selection time, error rates, recovery cost, and the relationship between local batching and attention. Such observations would test the transcript's workload scenarios instead of treating them as predictions already confirmed by a solver.

The native validation is hardware-specific. Controlled device-loss and process-termination checks do not simulate every driver fault or physical power loss. The unattended tests exercise production restore/file-processing paths but do not claim manual coverage of every common file or confirmation dialog. Likewise, a verified full-state bridge does not certify every possible upstream binary as compatible. These boundaries allow the demonstrated results to be used without enlarging them into unsupported guarantees.

10 Conclusion

The full 600-cell becomes tractable as a computational control problem when its pieces are separated into actual move orbits, orientations are represented by their true groups, and every macro retains its legal witness and complete collateral action. The original design transcript develops that construction and the accompanying reference makes its thirty-five seeds and setup machinery explicit. The conservative color-state bound explains the extraordinary scale relative to a documented solved high-dimensional benchmark while avoiding a universal difficulty claim.

The subsequent workbench turns those ideas into a tested native workflow with coherent state, recoverable operations, exact filtering, and measurable performance. Its present strength is verified macro-assisted control and responsive filtered or adaptive interaction. Economical human algorithms and continuous full-detail 1080p/30 FPS remain distinct research and engineering targets. The public report, original algorithm reference, synthetic replay companion, and bounded audit make these achievements and remaining questions independently inspectable.

Data availability and attribution

The repository includes this report and source, the unchanged 51-page algorithm reference, exact orbit words and selected tables, three synthetic solve records, the independent count audit, and the Puzzle theory supplement. Its introduction distinguishes historical reports from repeated checks. The private conversation is documented only by technical page ranges. Personal exchanges, account identifiers, databases, screenshots, diagnostics, third-party solve logs, and personal-history traces are excluded; no proprietary runtime assemblies are added.

Andrey Astrelin receives primary credit for the original MPUlt simulator and renderer; this mod retains its upstream notices. Nan Ma's solving methods and ivan216's projects are acknowledged as antecedents. AI-assisted assertions are assessed through explicit arguments, retained certificates and execution checks. This report is not a peer-reviewed publication.

A Complete moving-orbit census and controller costs

In Table 6, $r = |M| - 1$, k is stickers per piece, N is piece count, H is the local orientation group, $w = |W|$ is the raw seed length, $d_{\max}$ is the maximum retained guarded setup depth, and $C_{\max} = |W| + 2|R| + 2d_{\max}$ is the maximum star length for that tree. A superscript $*$ marks a raw seed with no collateral. Trivial H is denoted 1. Lengths count lab primitives and include the actual retained seed cancellation; they are not claimed globally minimal.

Table 6: All 35 moving orbits and retained controller costs.

Orbit	r	k	N	H	w	$d_{\max}$	$C_{\max}$
O00	1	2	1,200	C_2	2,182	20	2,258
O01*	2	1	3,600	1	2,190	12	2,236
O02	3	1	2,400	1	9,214	10	9,250
O03	3	2	3,600	C_2	1,094	10	1,130
O04*	4	1	7,200	1	546	8	578
O05*	4	1	7,200	1	546	8	578
O06	4	5	720	D_5	382	9	416
O07*	5	1	7,200	1	286	8	314
O08	5	2	3,600	C_2	1,534	7	1,558
O09	6	2	7,200	1	382	7	408
O10	7	1	7,200	1	766	7	790
O11	7	2	3,600	C_2	766	7	790
O12	8	1	7,200	1	766	6	788
O13	8	1	7,200	1	142	6	164
O14	8	2	7,200	1	1,534	6	1,556
O15	9	1	2,400	1	11,564	6	11,586
O16	9	2	7,200	1	286	6	308
O17	9	5	1,440	C_5	1,534	6	1,558
O18*	10	1	7,200	1	1,534	6	1,558
O19*	10	1	7,200	1	142	6	164
O20*	10	2	7,200	1	142	6	164
O21	11	1	7,200	1	94	6	116
O22	11	2	3,600	C_2	766	6	788
O23	12	2	7,200	1	46	6	68
O24	13	1	7,200	1	46	6	68
O25	13	2	3,600	C_2	46	6	68
O26	14	1	7,200	1	46	6	68
O27*	14	1	7,200	1	94	6	116
O28	14	5	1,440	C_5	3,070	6	3,092
O29	15	1	2,400	1	9,214	5	9,230
O30*	15	2	7,200	1	94	5	114
O31*	16	1	7,200	1	142	6	164
O32*	17	2	3,600	C_2	94	6	118
O33*	18	1	2,400	1	46	5	66
O34	19	20	120	A_5	12,286	5	12,304

The counts sum to 176,520 moving pieces and 259,200 moving stickers; adding 600 fixed centers gives 177,120 pieces and 259,800 stickers. Twelve raw seeds are pure and twenty-three have collateral. The retained triangular order is

```
006 000 017 015 002 022 021 008 023 009 029 025 013 011 010 034 028 026 024
016 014 012 003 033 032 031 030 027 020 019 018 007 005 004 001.
```

Every listed collateral edge points forward in this order. The table is regenerated from the immutable model's census, seed atlas and execution trees by the extended arithmetic checker. It summarizes those retained certificates and does not independently prove all their geometric transitions.

B Software contracts and reproducibility

The source locations below identify the inspected implementation and the appropriate verification layer. A file name is a route to evidence, not a claim that every possible execution path is tested. The application source reference is pinned in the bibliography; the expanded report's arithmetic scripts and results are published alongside its LaTeX source.

Table 7: Components, semantic contracts, and representative verification entry points.

Component	**Contract**	**Evidence / boundary**
`core.py`: Model	Asset hashes bind geometry, enumerations and controllers. Full primitive maps use one source-to-destination convention.	`test_reference_maps.py`: all 1,200 packed maps compared with a separate constructor; geometry is shared.
`core.py`: PuzzleState	Labels are a bijection; pieces stay intact and within orbit; exact, positional and color progress are distinguished.	`test_core.py`; complete-label comparisons, not screenshots alone.
`core.py`: Planner	Targets are nonbuffer members of the selected orbit; legal frames and full collateral are retained.	All-orbit seed/local-completion checks; unavailable residual corrections are rejected.
`core.py`: Filters	First-match Boolean rules are evaluated per piece; identity and position sets have different semantics.	`test_piece_filters.py`; parser bounds and native visibility tests address different failure layers.
`session.py`	Preview does not mutate state. Commit binds head/revision/hash and current protection; journal writes precede publication.	`test_core.py`, `test_crash.py`; process exits test transaction recovery, not hardware power failure.
`enhanced.py`	Scrambles, compositions, saved macros, branch reports and explicit timing use the same authoritative session.	Workflow regressions and native tools checks; no arbitrary user scripts are evaluated.
`log_io.py` and `mpult_log.py`	Complete detached replay and bounded decoding precede import; native token meanings are preserved.	`test_log_workflow.py`, `test_mpult_log.py`; interactive formats differ from research archives.
`server.py`	Authenticated loopback requests serialize state work; snapshot arrays describe one locked state.	`test_http_boundary.py`, `test_native_snapshot.py`; this is a local application API.

Continued on the next page.

Table 7, continued.

Component	Contract	Evidence / boundary
`engine_process.py`	Launch identity and health bind ownership even when launcher and interpreter PIDs differ; parent EOF cleans up.	24 lifecycle checks and three packaged-command checks in the retained validation.
`native_bridge.py`	A full slot bijection intertwines every translated generator; partial geometry cannot pass readiness.	`test_native_bridge.py` plus the actual runtime handshake; synthetic fixtures alone are insufficient.
`NativeSnapshot.cs` and `NativeHost.cs`	Validate one profile-bound snapshot before updating colors, styles, controls and history displays.	Native data, interaction and fresh-process reopen checks.
`NativePickingVisibility.cs`	Hidden meshes do not receive turns; a fresh visible projection precedes click hit-testing after sampled motion.	Native picking regression includes visible-frame and stale-axis cases.
`NativeRenderSubset.cs`	Temporary mesh scopes retain original identities and restore complete native arrays; frame policy is independent.	Native subset and frame-policy regressions, including direct and animated redraws.
`NativeRendererLifecycle.cs`	Dirty scheduling, handle/size guards and recognized device-reset recovery preserve control state.	Native device-recovery checks; no general proof over all graphics drivers.
`NativeFullRenderer.cs`	Only verified native instructions receive the reusable-buffer optimization; projection and order stay comparable.	Three full-detail poses and a selection/outline view match reference pixels; performance is workload-specific.
`packaging/`	Frozen engine, x86 host, resources and installer retain their tested identities; normal startup excludes developer fixtures.	Seven frozen-package checks and four isolated installer checks; external Managed DirectX remains a dependency.

B.1 Reproducing the report-specific arithmetic

The original `verify_color_bound.py` independently rebuilds the selected nine primitive actions, checks pure supports and relocations, reconstructs guarded position reachability, and evaluates the color bound. Its `results.json` records the specific input hashes and explicit omissions. The added `verify_extended_derivations.py` uses only declarative repository assets and the Python standard library. It recomputes the 35-row census/controller table, verifies the retained parent/depth structure and collateral order, checks the small-domain cycle and commutator identities, and calculates the structural upper count, word-ball lower bound, scramble-support inequality, and workload bound. Its `extended-results.json` records those results separately.

From the repository root, the added calculation is reproduced with:

```
python research/audit/verify_extended_derivations.py
```

No application session or native renderer is opened by this calculation. It does not independently regenerate clipping geometry, verify every oriented transition of all 35 setup trees, or replay the enormous primitive expansions. The three original synthetic replay commands remain documented inside the frozen companion ZIP. Their accepted endpoints and stage checks are preserved rather than replaced by newly generated solutions.

The separate command `pythonresearch/audit/verify_regular_geometry.py` records exact incidence, quaternion and mod-2 chain calculations in `regular-geometry-results.json`. Puzzle theory links the proof blueprint, strict verdict and review provenance [15].

B.2 Audit trail and publication scope

The new report-specific bounds are developments of the retained model and are not attributed retroactively to the original conversation or to Nan Ma's simplified solve. The original algorithm paper and curated reference ZIP remain byte-identical. Only the report, its LaTeX source, explanatory documentation, and bounded research supplements are expanded. The application release continues to be 0.2.4.

Checksums connect the PDF and archive to the published inventory, while source-control history identifies each documentation revision. This establishes byte identity and a reviewable lineage; it is not independent certification of authorship. The public materials retain no account-bearing conversation links, personal databases, original third-party histories, screenshots, raw local diagnostics, or credentials. Technical page ranges in the archival guide provide the necessary documentary connection without exposing those materials.

References

[1] Project archive. *600 Cell Solving Strategy*. Exported design transcript, 52 pages, 2026. Privately retained primary documentary source; public page-range guide supplied with this report.

[2] Project archive. *Full 600-Cell: 35-Orbit Algorithms*. Computational algorithm reference, 51 pages, 2026. Original supplied PDF and exact-word companion retained with the public research materials.

[3] Astrelin, A. Original announcement of the full 600-cell, 27 May 2011. Hypercubing archive, message 1743. https://archive.hypercubing.xyz/messages/1743.html Original MPUlt project and authorship: https://superliminal.com/andrey/mpu/

[4] Ma, N. Detailed simplified 600-cell solving method, 2012. Hypercubing archive, message 2443. https://archive.hypercubing.xyz/messages/2443.html

[5] Astrelin, A. *MC7D: Hall of Fame and puzzle documentation*. Original project site; accessed 7 September 2026. https://superliminal.com/andrey/mc7d/

[6] Gottlieb, M., and other participants. Completion account for the team $256 \times 256 \times 256$ solve, 2021. Speedsolving discussion, page 4, especially post 67. https://www.speedsolving.com/threads/256x256-rubiks-cube-solve-world-record-attempt.79808/page-4

[7] C600 Studio contributors. *0.2.4 Release Validation* and *MPUlt Runtime Provenance*. C600-MPUlt repository, 2026. https://github.com/KonomiYuzu01/C600-MPUlt/blob/main/tests/VALIDATION.md and https://github.com/KonomiYuzu01/C600-MPUlt/blob/main/docs/RUNTIME_PROVENANCE.md

[8] ivan216. *MPUlt*. Source project. https://github.com/ivan216/MPUlt

[9] ivan216. *mc7d-kb*. Source project. https://github.com/ivan216/mc7d-kb

[10] Microsoft. *DirectX End-User Runtimes (June 2010)*. Official runtime distribution. https://www.microsoft.com/en-us/download/details.aspx?id=8109

[11] C600 Studio contributors. Inspected application source, revision `7f1e250571e3`, 2026. https://github.com/KonomiYuzu01/C600-MPUlt/tree/7f1e250571e39c149434686641d1fbc6497c7905.

[12] SQLite project. *Write-Ahead Logging* and *PRAGMA synchronous*. Official documentation, accessed 7 September 2026. https://www.sqlite.org/wal.html and https://www.sqlite.org/pragma.html#pragma_synchronous.

[13] SQLite project. *Atomic Commit in SQLite*. Official documentation, accessed 7 September 2026. https://www.sqlite.org/atomiccommit.html.

[14] Amdahl, G. M. *Validity of the single processor approach to achieving large scale computing capabilities*. AFIPS Spring Joint Computer Conference, 1967, pp. 483–485. https://doi.org/10.1145/1465482.1465560. Author's paper: https://www.cs.cmu.edu/~18742/papers/Amdahl1967.pdf.

[15] C600 Studio project. *Puzzle Theory of the Full 600-Cell: Geometry, Topology, Group Actions, and Sound Optimization*. LaTeX supplement with geometry audit and Rethlas review, 7 September 2026; revision `d5dabe2c896f`. https://github.com/KonomiYuzu01/C600-MPUlt/blob/d5dabe2c896f1e39901f2cca707d4624004047d2/research/PUZZLE_THEORY.md.